

Notes: Chen, Goldstein, and Jiang (RFS, 2007)
Price Informativeness and Investment Sensitivity to Stock Price

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1 Big picture

- **Question.** Do managers learn from the private information embedded in their firms' stock prices when making corporate investment decisions?
- **Core hypothesis.** If stock prices contain information that managers do not already know, investment should be more sensitive to stock price when the stock price is more informative.
- **Main empirical idea.** The paper interacts Tobin's Q , the stock-price-based investment signal, with two measures of private information in price: price nonsynchronicity ($1 - R^2$) and probability of informed trading (PIN).
- **Main result.** Both $(1 - R^2) \times Q$ and $PIN \times Q$ are positive and statistically significant in investment regressions. Investment responds more to price when price appears to contain more private information.
- **Managerial learning channel.** The results are robust to controls for managerial information, including insider trading and earnings surprises. The interpretation is that at least part of the private information in stock prices is new to managers.
- **Alternative channel.** The paper also connects the result to the capital-constraints channel of Baker, Stein, and Wurgler. Both channels matter: price informativeness and financing constraints both help explain why investment and stock price move together.
- **Economic takeaway.** Financial markets are not just a valuation side show. By aggregating dispersed private information, stock prices can affect real corporate investment.

2 Incremental contribution of the paper

- **From stock prices predicting investment to managers learning from prices.** Earlier work documents that investment is correlated with stock price or Tobin's Q . This paper asks why the price-investment link is stronger in some firms than others.
- **A direct test of the feedback channel.** The paper tests the theory that managers observe stock prices, learn information from them, and use that information when choosing investment.
- **Two independent information measures.** The paper uses both price nonsynchronicity and PIN. These measures come from different literatures: one from return comovement, and one from market microstructure.
- **First use of PIN in a corporate investment setting.** The paper is one of the first to bring a market-microstructure measure of informed trading into corporate finance and real investment.
- **Managerial information controls.** By controlling for insider trading and earnings announcement surprises, the paper distinguishes information already held by managers from information that is new to managers.
- **Performance evidence.** The paper also shows that firms with more private information in price have better future operating performance, supporting the view that the information helps managers make better choices.

- **Clarifies multiple mechanisms behind investment-price sensitivity.** The paper does not deny the financing-constraints channel. Instead, it shows that the learning channel and the financing-constraints channel coexist.

3 Data and measurement

3.1 Sample construction

- The main sample is an unbalanced panel of Compustat firms from 1981 to 2001.
- Financial firms and utilities are excluded.
- The authors require at least \$10 million in book equity and at least 30 trading days in the year.
- The main sample contains 68,277 firm-year observations and 7,268 firms.
- The PIN sample is smaller, with 19,208 firm-year observations, because TAQ intraday trading data start in 1993.

Interpretation: The data combine accounting data, stock returns, market microstructure data, insider trading records, analyst coverage, and institutional ownership. The key advantage is that the paper can connect financial-market information production to real corporate investment.

3.2 Investment measures

The paper uses three dependent variables:

- **CAPXRND:** capital expenditures plus R&D, scaled by beginning-of-year assets.
- **CAPX:** capital expenditures, scaled by beginning-of-year assets.
- **CHGASSET:** percentage change in book assets.

Interpretation: CAPXRND is the broadest measure of organic investment, CAPX focuses on physical investment, and CHGASSET also captures acquisitions and divestitures. Similar results across these measures make the findings less dependent on one particular investment definition.

3.3 Stock price and cash-flow variables

- **Tobin's Q .** The main price variable is Tobin's Q , measured as market value of equity plus book assets minus book equity, all scaled by book assets.
- **Cash flow.** Cash flow equals net income before extraordinary items plus depreciation and amortization plus R&D, scaled by beginning-of-year assets.
- **Future returns.** The regressions include future market-adjusted returns to control for the possibility that investment reflects market timing or overvaluation.
- **Inverse assets.** The regressions include inverse assets to address common-scaling concerns because both investment and Q are scaled by assets.

Interpretation: The paper treats Q as the price signal available to managers. If the signal contains information that managers do not already know, firms should respond more strongly to Q .

3.4 Private information in stock price

The paper uses two measures of private information in price.

Price nonsynchronicity. The first measure is $1 - R^2$, where R^2 comes from the regression

$$r_{i,j,t} = \alpha_{i,0} + \alpha_{i,m}r_{m,t} + \alpha_{i,j}r_{j,t} + \varepsilon_{i,t}.$$

Here $r_{i,j,t}$ is the return of firm i in industry j , $r_{m,t}$ is the market return, and $r_{j,t}$ is the industry return. A higher $1 - R^2$ means that a larger share of stock-return variation is firm-specific rather than market- or industry-wide.

Probability of informed trading. The second measure is PIN, estimated from the Easley, Kiefer, and O'Hara market microstructure model. Conceptually,

$$PIN = \frac{\alpha\mu}{\alpha\mu + \epsilon_b + \epsilon_s},$$

where α is the probability of an information event, μ is the arrival rate of informed trades, and ϵ_b, ϵ_s are arrival rates of uninformed buy and sell orders.

Interpretation: The two information measures are complementary. $1 - R^2$ captures the outcome of firm-specific information entering price, while PIN captures the trading-process source of private information.

3.5 Managerial information and other controls

- **Insider trading.** The authors use the percentage of insider transactions among total transactions as a proxy for managerial private information.
- **Earnings surprise.** The authors use abnormal returns around quarterly earnings announcements as another proxy for information managers had before the market.
- **Analyst coverage.** Analyst coverage measures the number of analysts issuing forecasts or recommendations.
- **Capital constraints.** The authors use a four-variable Kaplan-Zingales score, $KZ4$, to proxy for equity dependence.
- **Size, diversification, and institutional ownership.** These variables address the concern that informativeness measures may proxy for other firm characteristics.

Interpretation: The controls are designed to separate the learning-from-price channel from other reasons why investment might be correlated with stock price, such as financing constraints, analyst attention, or firm size.

4 Models and empirical specifications

4.1 Baseline investment-price sensitivity model

The baseline empirical model is

$$I_{it} = \alpha_t + \eta_i + \beta_1 Q_{i,t-1} + \beta_2 INFO_{i,t-1} Q_{i,t-1} + \Gamma' CONTROL_{it} + \varepsilon_{it}.$$

- I_{it} is investment.
- $Q_{i,t-1}$ is the lagged stock-price signal.
- $INFO_{i,t-1}$ is either $1 - R^2$ or PIN.
- β_1 measures investment sensitivity to price for the median-information firm.
- β_2 is the key coefficient: it asks whether investment is more sensitive to price when price contains more private information.
- Firm and year fixed effects are included.
- Standard errors are clustered by firm.

Interpretation: The learning hypothesis predicts $\beta_2 > 0$. A positive interaction means that the same stock price signal matters more for investment when the price has more firm-specific or informed-trading content.

4.2 Managerial information specification

To test whether the stock-price information is new to managers, the authors add interactions of managerial information proxies with Q :

$$I_{it} = \alpha_t + \eta_i + \beta_1 Q_{i,t-1} + \beta_2 INFO_{i,t-1} Q_{i,t-1} + \beta_3 MANAGER_{i,t-1} Q_{i,t-1} + \Gamma' CONTROL_{it} + \varepsilon_{it}.$$

- $MANAGER$ is proxied by insider trading or earnings surprises.
- If managers already know the relevant information, the stock-price sensitivity should be lower when managerial information is high.
- The important robustness question is whether $INFO \times Q$ remains positive after these controls.

Interpretation: The paper finds that managerial information proxies tend to reduce investment-price sensitivity, while $(1 - R^2) \times Q$ and $PIN \times Q$ remain positive. This supports the interpretation that price contains information not already known by managers.

4.3 Analyst coverage specification

The paper also estimates models with $ANALYST \times Q$. Analyst coverage can either make price more useful to managers or mostly transmit information from managers to investors.

Interpretation: The coefficient on $ANALYST \times Q$ is negative. The authors interpret this as consistent with analysts mainly transmitting information that managers already know, especially in the pre-Reg FD sample period.

4.4 Capital constraints and size sorts

The paper estimates the baseline model separately by:

- quintiles of the Kaplan-Zingales $KZ4$ financing constraint index; and
- quintiles of market capitalization.

Interpretation: The sorting results show that the learning channel is not merely a disguised financing-constraints channel or size effect. Private information in price continues to affect investment-price sensitivity within these subsamples.

4.5 Future performance regressions

The authors also regress future operating performance on scores based on price informativeness:

$$Performance_{i,t:t+3} = \alpha_t + \eta_i + \theta Score_{it} + \Gamma' X_{it} + \varepsilon_{it}.$$

- Performance measures are ROA, sales growth, and asset turnover.
- The score is based on $1 - R^2$, or a weighted combination of $1 - R^2$ and PIN.

Interpretation: If private information in price helps managers make better investment decisions, firms with more informative prices should perform better later. The paper finds this positive relation.

5 Identification and empirical design

5.1 What the design identifies

- The paper identifies whether the stock-price-investment relation is stronger when stock prices appear to contain more private information.
- The empirical design is primarily panel-regression based, using firm and year fixed effects and many controls.
- The strongest interpretation is that managers learn from prices, but the paper acknowledges that this is a joint test of the information measures and the learning hypothesis.

Interpretation: This is not a randomized experiment. The paper is persuasive because many different empirical patterns point in the same direction: two independent information measures, managerial information controls, analyst coverage, performance outcomes, and robustness checks.

5.2 Why $1 - R^2$ and PIN are central

- $1 - R^2$ is high when stock returns are less synchronized with market and industry returns.
- PIN is high when order-flow patterns suggest a higher probability of informed trading.
- Both are interpreted as measures of private information in price.

Interpretation: The central identifying logic is cross-sectional and time-series heterogeneity in price informativeness. If prices with more private information generate stronger investment-price sensitivity, that is consistent with feedback from prices to real decisions.

5.3 Main threats to interpretation

- **Measurement interpretation.** Price nonsynchronicity and PIN may capture something other than private information, such as noise or firm characteristics.
- **Firm size.** Both information measures are correlated with size, and size may independently affect investment-price sensitivity.
- **Financing constraints.** The price-investment relation may reflect the ability to raise capital rather than learning.
- **Managerial information.** The price may contain information managers already know, in which case investment would not be reacting to price itself.

Interpretation: The paper addresses these concerns with controls, subsample analysis, alternative specifications, and performance tests. The evidence is not purely causal, but the robustness exercises substantially narrow the set of alternative explanations.

6 Tables and figures

6.1 Table 1: Variable definitions and summary statistics

Read:

- Panel A defines the key variables: investment measures, Tobin's Q , price nonsynchronicity, PIN, cash flow, managerial-information proxies, analyst coverage, and future performance measures.
- Panel B shows the scale of the data. The main sample has 68,277 observations; the PIN sample has 19,208 observations.
- Mean $1 - R^2$ is 0.83, indicating that much of firm return variation is firm-specific.
- Mean PIN is 0.21, close to estimates in the microstructure literature.

Economic message: Table 1 establishes the paper's measurement system. The paper is not only about whether investment follows prices; it is about whether price information quality changes the strength of that relation.

Table 1
Variable definitions and summary statistics

Variable	Definition
Panel A: Definitions	
<i>CAPXRND</i>	Capital expenditure plus R&D scaled by beginning-of-year assets (%)
<i>CAPY</i>	Capital expenditure scaled by beginning-of-year assets (%)
<i>CHGASSET</i>	Change in assets scaled by beginning-of-year assets (%)
<i>Q</i>	Market value of equity plus book value of assets minus book value of equity, scaled by book value of assets
$1 - R^2$	One minus R^2 from regressing daily return on market and industry index over ye
<i>PIN</i>	PIN measure per Easley et al. (1996)
<i>CF</i>	Net income before extraordinary item + depreciation and amortization expenses
	R&D expenses, scaled by lagged assets
<i>RET</i>	Value-weighted market return adjusted firm return for next three years
<i>ASSET</i>	Total book value of assets in \$billions
<i>INV_AST</i>	Inverse of <i>ASSET</i>
<i>INSIDER</i>	Number of transactions by insiders scaled by total number of transactions record in TAQ
<i>KZ4</i>	Four-variable KZ score (excluding Q) per Kaplan and Zingales (1997)
<i>ERC</i>	Average of the absolute stock returns over the four quarterly earnings announcer periods (day -1 to day 1) (in %)
<i>SIZE</i>	Market capitalization (\$million)
<i>SALES</i>	Total sales revenues (\$million)
<i>HERFINDAHL</i>	Herfindahl index of sales based on firms segment reports
<i>ANALYST</i>	Number of analysts issuing forecasts or recommendations for the firm
<i>INSTITUTION</i>	Percentage of shares held by institutional investors
<i>ROA</i>	Operating earnings (i.e., earnings before interest, taxes, depreciation, and amortization) as a percentage of market value of assets, which is the sum of market value of equity and book value of debt (in %)
<i>Sales growth</i>	Annual growth rate in sales revenues (%)
<i>Asset turnover</i>	Sales revenue divided by total asset values (%)

Variable	Number of observations	MEAN	SD	5%	25%	50%	75%	95%
Panel B: Summary statistics								
<i>CAPXRND</i>	68277	14.81	17.53	1.22	4.78	9.61	17.85	45
<i>CAPY</i>	68277	9.82	11.76	0.82	3.29	6.28	11.48	31
<i>CHGASSET</i>	68274	29.22	79.08	-18.77	-0.52	9.51	26.59	137
<i>Q</i>	64783	1.81	1.48	0.75	1.00	1.31	1.98	4
$1 - R^2$	68277	0.83	0.23	0.27	0.79	0.92	0.98	1
<i>PIN</i>	19208	0.21	0.08	0.09	0.16	0.21	0.26	0
<i>CF</i>	68276	0.13	0.16	-0.09	0.06	0.12	0.19	0
<i>RET</i>	68277	0.02	0.81	-0.88	-0.50	-0.12	0.28	1
<i>ASSET</i>	68277	1.30	3.90	0.02	0.06	0.16	0.62	6
<i>INSIDER</i>	25412	2.78	7.72	0.00	0.00	0.34	1.78	13
<i>KZ4</i>	68176	0.01	1.65	-2.44	-0.60	0.18	1.02	2
<i>ERC</i>	56029	5.13	3.58	1.24	2.61	4.20	6.61	12
<i>SIZE</i>	68277	1650.95	9726.13	12.36	44.27	137.58	576.76	5625
<i>SALES</i>	68277	1263.97	3550.92	12.75	59.20	179.71	681.90	6296
<i>HERFINDAHL</i>	53501	0.94	0.16	0.51	1.00	1.00	1.00	1
<i>ANALYST</i>	58618	5.29	6.53	0.00	1.00	3.00	7.00	19
<i>INSTITUTION</i>	32037	37.6	21.9	5.16	19.5	35.9	54.5	75
<i>ROA</i>	64708	13.76	10.00	-3.29	8.42	13.77	19.19	30
<i>Sales growth</i>	59965	16.34	40.51	-26.55	-1.44	9.04	23.21	80
<i>Asset turnover</i>	68277	149.23	104.41	25.59	82.68	129.68	186.36	347

(b) Table 1, Panel B: summary statistics

(a) Table 1, Panel A: variable definitions

Table 1: Variable definitions and summary statistics.

6.2 Table 2: Relation between investment-price sensitivity and private information measures

Read:

- Table 2 is the core baseline result.
- Across investment measures, Q is positively related to investment.
- The interaction $(1 - R^2) \times Q$ is positive and significant in most specifications.
- The interaction $PIN \times Q$ is also positive and significant.
- When both information measures enter together, both retain positive explanatory power, suggesting they capture distinct parts of price informativeness.
- $(1 - R^2) \times CF$ is negative, suggesting that when price is more informative, investment relies less on cash flow.

Economic message: The baseline table supports the paper’s main learning channel: managers respond more to stock prices when prices contain more private information.

Table 2
Relation between investment-price sensitivity and private information measures

Dependent variable	CAPXRND			CAPX		
	1	2	3	4	5	6
<i>Q</i>	3.52*	2.18*	1.95*	2.20*	1.37*	1.0
	0.08	0.10	0.12	0.06	0.06	0.0
$(1-R^2)*Q$	3.13*	—	1.80*	1.42*	—	0.4
	0.28		0.42	0.20		0.2
<i>PIN*Q</i>	—	4.21*	3.37*	1.92*	1.7	1.7
	—	1.04	1.11	0.67	0.6	0.6
<i>CF</i>	—	—	17.68*	—	7.6	7.6
	—	—	1.33	—	—	0.6
$(1-R^2)*CF$	—	—	-29.09*	—	—	-17.4
	—	—	6.33	—	—	3.9
<i>PIN*CF</i>	—	—	14.89	—	10.0	10.0
	—	—	14.37	—	7.0	7.0
<i>RET</i>	—	—	-0.21***	—	—	-0.4
	—	—	0.11	—	—	0.0
<i>INV_AST</i>	—	—	-0.05	—	-0.0	-0.0
	—	—	0.03	—	—	0.0
$1-R^2$	-7.20*	—	2.78*	-5.62*	—	1.0
	0.56	—	1.04	0.46	—	0.7
<i>PIN</i>	—	-3.31***	-5.31**	—	-1.09	-2.4
	—	1.98	2.21	—	0.76	1.5
Adjusted R^2	0.54	0.54	0.57	0.44	0.44	0.4
Within R^2	0.10	0.08	0.14	0.07	0.06	0.1

Definitions of all variables are listed in Table 1 Panel A. Dependent variables (*CAPXRND*, *CAPX*, and *CHGASSET*) beginning of the year. Both firm- and year-fixed effects are included. Coefficient estimates are shown in bold and their : adjust for both heteroskedasticity and within correlation clustered by firm. Number of observations is 64,782 for Co *, **, and *** indicate a two-tailed test significance level of less than 1%, 5%, and 10%, respectively.

(a) Table 2, CAPXRND and CAPX columns

Table 2
Relation between invc

Dependent variable		CHGASSET		
		7	8	9
<i>Q</i>	8*	20.26*	15.47*	10.56*
	6	0.53	0.69	0.71
$(1-R^2)*Q$	5*	14.79*	—	2.70
	5	1.72	—	2.65
<i>PIN*Q</i>	0**	—	19.83*	20.33*
	7	—	6.36	6.28
<i>CF</i>	7*	—	—	135.88*
	7	—	—	6.78
$(1-R^2)*CF$	3*	—	—	-181.35*
	3	—	—	38.39
<i>PIN*CF</i>	1	—	—	-32.11
	7	—	—	74.97
<i>RET</i>	4*	—	—	-3.58*
	7	—	—	0.62
<i>INV_AST</i>	8*	—	—	-2.33*
	1	—	—	0.15
$1-R^2$	2	-23.09*	—	37.22*
	7	3.23	—	6.64
<i>PIN</i>	9	—	-9.28	-11.68
	7	—	11.58	11.96
Adjusted R^2	6	0.20	0.00	0.17
Within R^2	0	0.12	0.13	0.28

Definitions of all variables are expressed as percentage points of book assets at the beginning of the year. Standard errors are displayed right below. Standard errors adjust for both heteroskedasticity and within correlation clustered by firm. Number of observations is 64,782 for Co *, **, and *** indicate a two-tailed test significance level of less than 1%, 5%, and 10%, respectively.

(b) Table 2, CHGASSET columns with row labels

Table 2: Investment-price sensitivity and private information measures.

6.3 Table 3: Managerial information and analyst coverage

Read:

- Table 3 controls for information that managers may already possess.
- Insider trading intensity and earnings surprises are used as proxies for managerial private information.
- The interaction of earnings surprise with Q is negative and significant, consistent with managers relying less on price when they already have more information.
- The interactions $(1 - R^2) \times Q$ and $PIN \times Q$ remain positive and significant even after these controls.
- Analyst coverage lowers investment-price sensitivity, consistent with analysts mainly transmitting information already known to managers during the pre-Reg FD period.

Economic message: Table 3 is important because it moves the interpretation from “prices contain information” to “prices contain information managers do not already know.”

6.4 Table 4: Financial constraints and firm size

Read:

- Panel A sorts observations by the Kaplan-Zingales financing-constraint index.

- Investment-price sensitivity rises with financial constraints, confirming the Baker-Stein-Wurgler financing channel.
- The information interactions remain positive across constraint quintiles.
- Panel B sorts firms by market size. Investment-price sensitivity declines with size, but private information interactions remain positive across size quintiles.

Economic message: Table 4 shows that learning from prices and financing constraints are separate mechanisms. The price signal matters partly because it informs managers and partly because prices affect financing capacity.

Table 3
Controlling for managerial information and analyst coverage

	1	2	3	4
	INSIDER	PIN*	ERC	ANALYST
Q	1.98*	1.97*	2.04*	1.98*
	0.12	0.12	0.12	0.12
$MANAGER*Q$	-6.57	—	-2.53**	—
	6.73	—	1.02	—
$ANALYST*Q$	—	—	—	-0.57*
	—	—	—	0.08
$(1 - R^2)*Q$	1.84*	1.84*	1.79*	0.87*
	0.42	0.42	0.41	0.33
$PIN*Q$	3.16*	3.22*	3.30*	1.40**
	1.08	1.06	1.13	0.70
Number of observations	19130	19130	18722	19208
Adjusted R^2	0.58	0.58	0.58	0.58
Within R^2	0.14	0.14	0.15	0.15

Definitions of all variables are listed in Table 1 Panel A. The dependent variable is $CAPXRND$. The variable $MANAGER$ represents the manager's relative information advantage over the market, which is proxied, from Columns 1–3, by $INSIDER$ (the percentage of insider transactions to total number of transactions), PIN^* (the PIN measure adjusted to exclude insider trading from informed trading, and ERC (the average of the absolute market-model abnormal stock returns around the two-day windows of the four quarterly earnings announcement dates in the previous year). $ANALYST$ is the number of Zacks analysts covering the firm during the previous year, in logarithms. Both firm- and year-fixed effects are included. Shown are coefficient estimates for Q , $MANAGER*Q$, $(1 - R^2)*Q$, and $PIN*Q$ (in bold print) and their standard errors (displayed right below). Standard errors adjust for both heteroskedasticity and

(a) Table 3: managerial information and analyst coverage

Table 4
Effects of financial constraints and size

	Q1	Q2	Q3	Q4	Q5
Panel A: Quintiles formed by KZ					
KZ4 Quintiles					
Q	0.42*	0.44*	1.06*	1.94*	3.17*
	0.14	0.17	0.25	0.37	0.46
$(1 - R^2)*Q$	2.49*	1.97*	1.99**	3.82*	1.71
	0.52	0.57	0.97	1.17	1.05
$PIN*Q$	2.63***	2.10	3.80**	4.80	1.80
	1.52	1.41	1.55	3.86	3.19
Number of observations	3391	4324	4146	3968	3358
Adjusted R^2	0.28	0.28	0.23	0.18	0.15
Within R^2	0.13	0.07	0.08	0.08	0.06
Panel B: Quintiles formed by market size					
SIZE Quintiles					
Q	4.22*	5.47*	4.08*	2.31*	1.59*
	0.55	0.42	0.28	0.21	0.29
$(1 - R^2)*Q$	0.60	0.59	1.49***	1.97**	1.44*
	1.65	0.93	0.88	0.75	0.57
$PIN*Q$	0.62	1.58	0.73	0.54	4.98*
	3.29	3.14	1.65	1.90	1.74
Number of observations	2989	3557	3863	4056	4743
Adjusted R^2	0.49	0.67	0.66	0.69	0.71
Within R^2	0.07	0.16	0.20	0.19	0.29

Definitions of all variables are listed in Table 1 Panel A. All observations are sorted into five subsamples depending on the quintile in which a firm's $KZ4$ score (Panel A) or market size (Panel B) falls during the previous year. $KZ4$ is used to proxy for a firm's financing constraints. Regressions of $CAPXRND$ on Q , $(1 - R^2)*Q$, $PIN*Q$, and the same control variables as in Table 3 are estimated in each quintile. We estimate the five-quintile equations simultaneously with both year- and firm-fixed effects. Shown are the coefficient estimates for Q , $(1 - R^2)*Q$ and $PIN*Q$ (in bold font) and their standard errors (displayed right below). Standard errors adjust for both heteroskedasticity and within correlation clustered by firm.

(b) Table 4: financial constraints and size

Tables 3 and 4: Controls and subsample evidence.

6.5 Table 5: Future operating performance

Read:

- Table 5 asks whether firms with more informative prices perform better later.
- The dependent variables are future ROA, sales growth, and asset turnover.
- Price informativeness scores are positively related to all three performance measures.
- The result holds when the score is based only on $1 - R^2$ and when it combines $1 - R^2$ and PIN.

Economic message: Table 5 supports the interpretation that the information in stock prices is useful rather than pure noise. If the measures captured only noise, they should not positively predict future operating performance.

6.6 Table 6: Portfolio approach

Read:

- Table 6 sorts firm-year observations into quintiles by $1 - R^2$ or PIN.
- Investment-price sensitivity is lowest in the lowest-information quintile and highest in the highest-information quintile.
- The increase is especially monotonic for PIN quintiles.
- Investment-cash-flow sensitivity tends to decline as price informativeness increases.

Economic message: Table 6 shows that the main result is not driven by a few extreme observations. The relation between informativeness and investment-price sensitivity is broad-based across the distribution.

Table 5
Regression of future performance on investment-price sensitivity scores

	1	2	3	4	5	6
Dependent variable	ROA	ROA	Sales Growth	Sales Growth	Asset Turnover	Asset Turnover
<i>SCORE</i> based on	<i>I-R2</i>	<i>I-R2&PIN</i>	<i>I-R2</i>	<i>I-R2&PIN</i>	<i>I-R2</i>	<i>I-R2&PIN</i>
<i>SCORE</i>	1.51*	0.42***	5.96*	9.59*	11.38*	5.11*
	0.17	0.23	0.78	1.29	1.03	1.27
<i>SALES</i>	-0.83*	-0.95*	-3.09*	-3.09*	-7.79*	-17.15*
	0.06	0.14	0.22	0.22	0.45	1.08
<i>KZA</i>	0.07***	0.43*	-2.65*	-2.20*	-2.32*	0.28
	0.04	0.06	0.22	0.49	0.24	0.33
<i>HERFINDAHL</i>	2.02*	1.03*	7.72*	11.60*	19.48*	11.14*
	0.24	0.32	1.08	1.88	1.65	2.19
<i>Q</i>	-0.56*	-0.20*	-0.44**	-1.96*	-1.49*	-2.20*
	0.03	0.04	0.22	0.36	0.19	0.25
Number of observations	46304	11315	46393	11351	46409	11354
Adjusted R ²	0.68	0.78	0.34	0.29	0.86	0.91
Within R ²	0.03	0.03	0.02	0.03	0.04	0.15

Definitions of all variables are listed in Table 1 Panel A. The dependent variable is *ROA* in columns 1 and 2. *Sales Growth* in columns 3 and 4, and *Asset Turnover* in columns 5 and 6. *ROA* is calculated as the percentage of operating earnings to firms' total market value of assets (sum of market value of equity and book value of liabilities). *Sales Growth* is the annual growth rate in sales revenues. *Asset Turnover* is the percentage ratio of sales revenue to total assets. All dependent variables are averages over the three-year periods after year t . *SCORE* is a variable between 0 and 1, representing the percentile of the "learning score" in the sample. The score based on $I - R2$ is the percentage ranking of the observation's $I - R2$ in the sample. The score based on both $I - R2$ and PIN is calculated as the percentage ranking of a weighted average of these two values, with the weights being the coefficient estimates on $I - R2$ and PIN from regressions in Table 3. *SALES* is the sales revenue in year $t - 1$. Both firm- and year-fixed effects are included. Coefficient estimates are printed in bold and their standard errors are displayed right below. Standard errors adjust for both heteroskedasticity and within correlation clustered by firm.

*, **, and *** indicate a two-tailed test significance level of less than 1, 5, and 10%, respectively.

(a) Table 5: future operating performance

Table 6
Sensitivity to alternative specification: the portfolio approach

	Q1	Q2	Q3	Q4	Q5
Panel A: Quintiles formed by $1 - R2$					
<i>Q</i>	1.70*	2.63*	3.35*	3.24*	3.92*
	0.19	0.19	0.24	0.23	0.36
<i>CF</i>	31.64*	26.88*	24.11*	25.80*	25.87*
	2.24	2.05	2.06	2.01	2.23
<i>RET</i>	-0.41***	-0.11	0.11	-0.13	-0.08
	0.23	0.20	0.22	0.15	0.18
<i>INV_AST</i>	0.25*	0.17*	0.12*	0.07*	0.04*
	0.04	0.02	0.01	0.01	0.01
Number of observations	9602	9610	9613	9610	9598
Adjusted R ²	0.28	0.24	0.21	0.20	0.20
Panel B: Quintiles formed by PIN					
<i>Q</i>	1.55*	2.69*	3.33*	3.67*	5.26*
	0.22	0.23	0.34	0.33	0.50
<i>CF</i>	29.32*	24.34*	22.23*	19.44*	18.79*
	3.13	3.05	3.41	2.73	3.64
<i>RET</i>	0.45	0.64**	0.78***	0.31	0.72*
	0.30	0.32	0.41	0.29	0.35
<i>INV_AST</i>	0.49*	0.32*	0.22*	0.12*	0.07*
	0.09	0.04	0.03	0.02	0.02
Number of observations	3581	3310	3160	2951	2721
Adjusted R ²	0.30	0.28	0.22	0.23	0.26

Definitions of all variables are listed in Table 1 Panel A. Each firm-year observation is assigned into quintiles by its $I - R2$ value in Panel A and by its PIN value in Panel B, with the lowest value in quintile and the highest value in quintile 5. The dependent variable is *CAPXND*. Shown below are coefficient estimates for each quintile, with standard errors shown below. Year-fixed effects are included. Standard errors adjust for both heteroskedasticity and within correlation clustered by firm.

*, **, and *** indicate a two-tailed test significance level of less than 1, 5, and 10%, respectively.

(b) Table 6: portfolio approach

Tables 5 and 6: Performance and portfolio robustness evidence.

6.7 Table 7: Fama-MacBeth cross-sectional approach

Read:

- Table 7 estimates cross-sectional regressions year by year and averages the coefficients.
- The purpose is to distinguish cross-firm patterns from within-firm panel variation.
- $PIN \times Q$ remains positive and significant across the major specifications.
- $(1 - R^2) \times Q$ is also positive in most specifications.

Economic message: Table 7 shows that the main pattern is not an artifact of pooled panel estimation. More informative prices are associated with stronger investment-price sensitivity across firms.

6.8 Table 8: Institutional investors, diversification, and pure-play firms

Read:

- Table 8 adds controls for institutional ownership and diversification.
- Institutional holdings reduce investment-price sensitivity, consistent with their correlation with analyst coverage and firm size.
- Diversified firms have lower investment-price sensitivity, consistent with stock prices being less informative about individual segment investment opportunities.
- In the pure-play sample, $(1 - R^2) \times Q$ and $PIN \times Q$ remain positive and significant.

Economic message: Table 8 reinforces that the learning result is not simply a diversification, institutional-ownership, or segment-structure artifact.

Table 7
Sensitivity to cross-sectional correlations: the Fama-Macbeth approach

Dependent variable	1	2	3	4	5	6
	<i>CAPXRND</i>	<i>CAPX</i>	<i>CHGASSET</i>	<i>CAPXRND</i>	<i>CAPX</i>	<i>CHGASSET</i>
<i>Q</i>	3.55*	1.55*	12.95*	2.74*	0.37**	7.44*
	0.22	0.14	0.72	0.23	0.16	0.48
$(1-R^2) \times Q$	3.91*	0.33	10.76*	4.16*	0.33	-1.62
	0.68	0.49	1.50	0.69	0.46	2.33
<i>PIN</i> × <i>Q</i>	—	—	—	3.93**	3.01**	34.42*
	—	—	—	1.90	1.42	6.43
<i>CF</i>	29.91*	16.17*	65.63*	24.86*	10.67*	68.55*
	1.38	1.03	4.89	1.76	1.23	5.64
$(1-R^2) \times CF$	-17.83*	-0.23	25.65	-30.95*	-12.83**	23.82
	5.09	4.37	18.71	9.03	6.48	28.98
<i>PIN</i> × <i>CF</i>	—	—	—	-3.97	4.49	46.65
	—	—	—	19.21	14.41	66.52
<i>RET</i>	-0.90*	-0.99*	-3.06*	0.03	-0.62	-1.67*
	0.10	0.09	0.34	0.17	0.12	0.52
<i>INV_AST</i>	0.04*	-0.04*	-0.20*	0.12*	-0.06*	-0.57*
	0.01	0.00	0.02	0.01	0.01	0.03
$1 - R^2$	-2.10*	-0.38	-9.32*	-2.09	0.11	7.18**
	0.77	0.72	2.23	1.26	0.84	3.83
<i>PIN</i>	—	—	—	-1.46	2.86	-6.78
	—	—	—	3.12	2.49	11.64

Definitions of all variables are listed in Table 1 Panel A. Regressions are estimated for each year. Reported coefficient estimates are the averages of yearly estimates. Standard errors are estimated with clustered controlled bootstrapping. Number of observations is 64,782 in Columns 1–3 and 19,208 in Columns 4–6.

(a) Table 7: Fama-MacBeth approach

Table 8
Sensitivity to other factors affecting investment-price sensitivity

	1	2	3
	<i>INSTITUTION</i>	<i>HERFINDAHL</i>	<i>PURE PLAY</i>
<i>CONTROL</i>			
<i>Q</i>	2.02*	2.06*	2.03*
	0.13	0.14	0.15
$(1-R^2) \times Q$	1.14*	2.15*	1.80*
	0.36	0.47	0.53
<i>PIN</i> × <i>Q</i>	1.49**	3.41*	4.49*
	0.76	1.23	1.38
<i>CONTROL</i> × <i>Q</i>	-2.72*	0.57***	—
	0.37	0.30	—
Number of observations	15,635	15,213	11,518
Adjusted R ²	0.60	0.51	0.49
Within R ²	0.17	0.15	0.14

Definitions of all variables are listed in Table 1 Panel A. The dependent variable is *CAPXRND*. In Columns 1 and 2, the dependent variables are those in Table 3 plus the additional *CONTROL* variable: *INSTITUTION* (measured as the average percentage of shares held by institutional investors in year $t-1$) and *HERFINDAHL* (the Herfindahl index of firms' reported business segments based on previous-year sales). In Column 3, the same regression (without the extra *CONTROL* variable) is performed on "pure play" firms (i.e., firms that have only one reported business segment). Both firm- and year-fixed effects are included. Coefficient estimates for *Q*, $(1 - R^2) \times Q$, *PIN* × *Q* and *CONTROL* × *Q* are printed in bold and their standard errors are displayed right below. Standard errors adjust for both heteroskedasticity and within correlation clustered by firm.

*, **, and *** indicate a two-tailed test significance level of less than 1, 5, and 10%, respectively.

(b) Table 8: other factors and pure-play firms

Tables 7 and 8: Cross-sectional and additional robustness evidence.

7 Short summary

- The paper studies whether managers learn from private information in stock prices when making investment decisions.
- It measures private information using price nonsynchronicity $(1 - R^2)$ and *PIN*.
- The main regression interacts these information measures with Tobin's *Q*.
- Both $(1 - R^2) \times Q$ and $PIN \times Q$ are positive and significant.
- The result holds across multiple investment measures.
- It survives controls for cash flow, future returns, inverse assets, managerial information, analyst coverage, financial constraints, firm size, institutional holdings, and diversification.
- Managerial information proxies reduce investment-price sensitivity, consistent with managers relying less on price when they already know more.

- Analyst coverage also reduces investment-price sensitivity, consistent with analysts transmitting information from managers to markets rather than producing new information for managers.
- Firms with more private information in price have better future operating performance.
- The paper’s central message is that stock prices can guide real investment decisions by aggregating information that managers do not otherwise possess.

8 Conclusion

8.1 Core conclusion

Chen, Goldstein, and Jiang show that corporate investment is more sensitive to stock prices when stock prices contain more private information. This pattern is consistent with a feedback channel from financial markets to real corporate decisions: managers observe prices, infer information from them, and incorporate that information into investment choices.

8.2 Economic intuition

The intuition is simple. A stock price combines public information, private information produced by market participants, and noise. If the price reflects information that managers do not already know, then it should be useful for investment decisions. Therefore, a more informative price should generate a stronger investment response. The positive coefficients on $(1 - R^2) \times Q$ and $PIN \times Q$ are exactly this pattern.

8.3 Incremental contribution revisited

The paper advances the investment-price literature by distinguishing two mechanisms. The first is the financing-constraints channel: a high stock price can relax financing constraints. The second is the learning channel: a high or low stock price can reveal information about fundamentals. The paper shows that both channels matter, and it provides evidence that the learning channel is economically important.

8.4 Implications for financial markets

The findings imply that financial markets affect the real economy not only by allocating capital across investors but also by transmitting information to firms. This gives market transparency, informed trading, analyst activity, and market microstructure direct relevance for real corporate policy.

8.5 Limitations

The paper relies on observational panel regressions and on proxies for private information. Price nonsynchronicity and PIN are well-motivated, but they may still capture other firm characteristics. The authors acknowledge that the results are a joint test of the information measures and the

managerial-learning hypothesis. The robustness checks make alternative explanations less plausible, but they do not eliminate every possible omitted variable concern.

8.6 Takeaway

The enduring takeaway is that stock prices can be decision inputs for managers. When prices aggregate private information, they do more than summarize value; they help shape real investment.